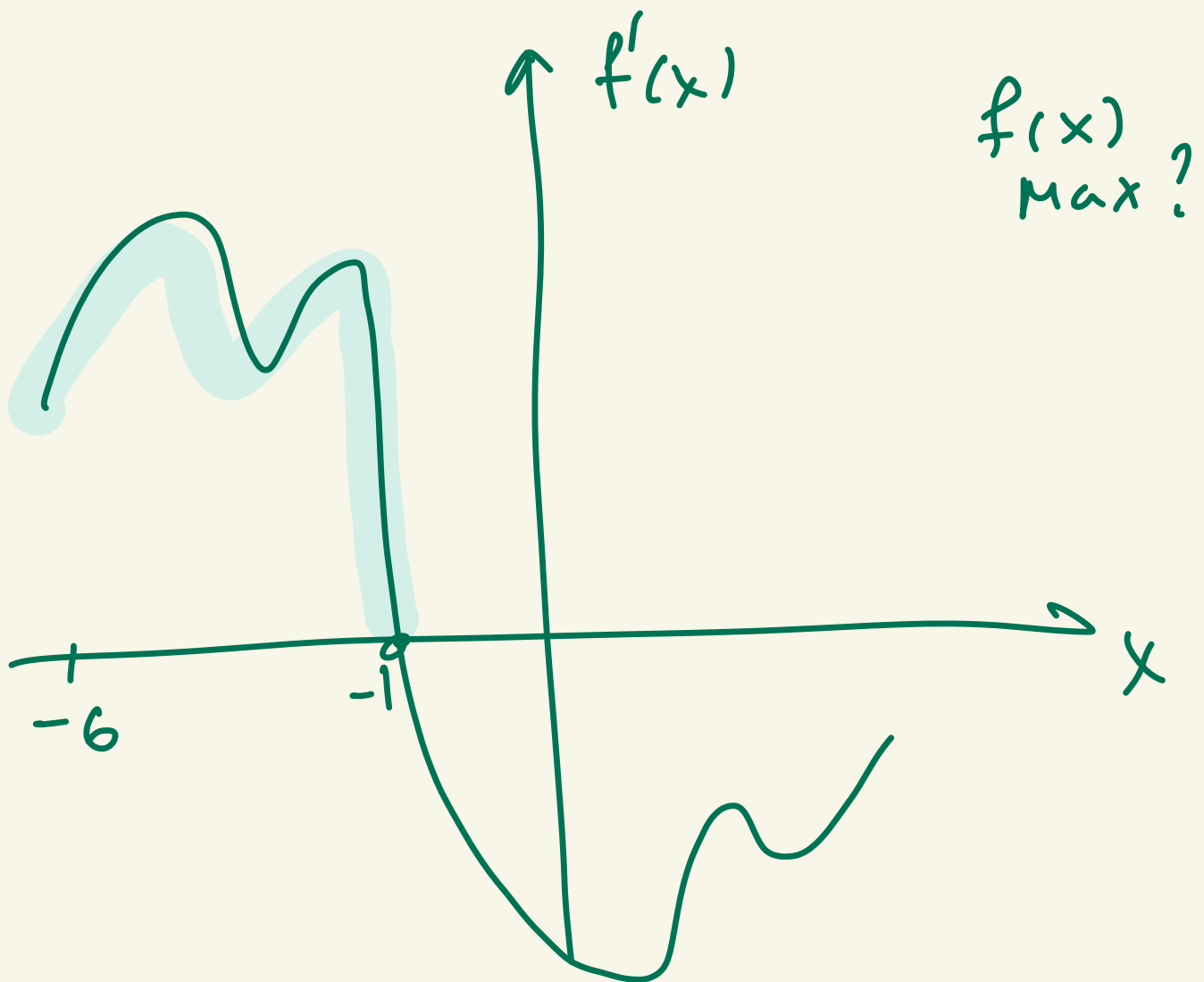






$$A \xleftarrow{165} B$$

$$\square \rightarrow v_n = v_t = 4 \text{ km} / \text{s}$$

$$\frac{92}{4} = t_n = 23$$

$$t_a = t_n - 1 = 22 \text{ s}$$

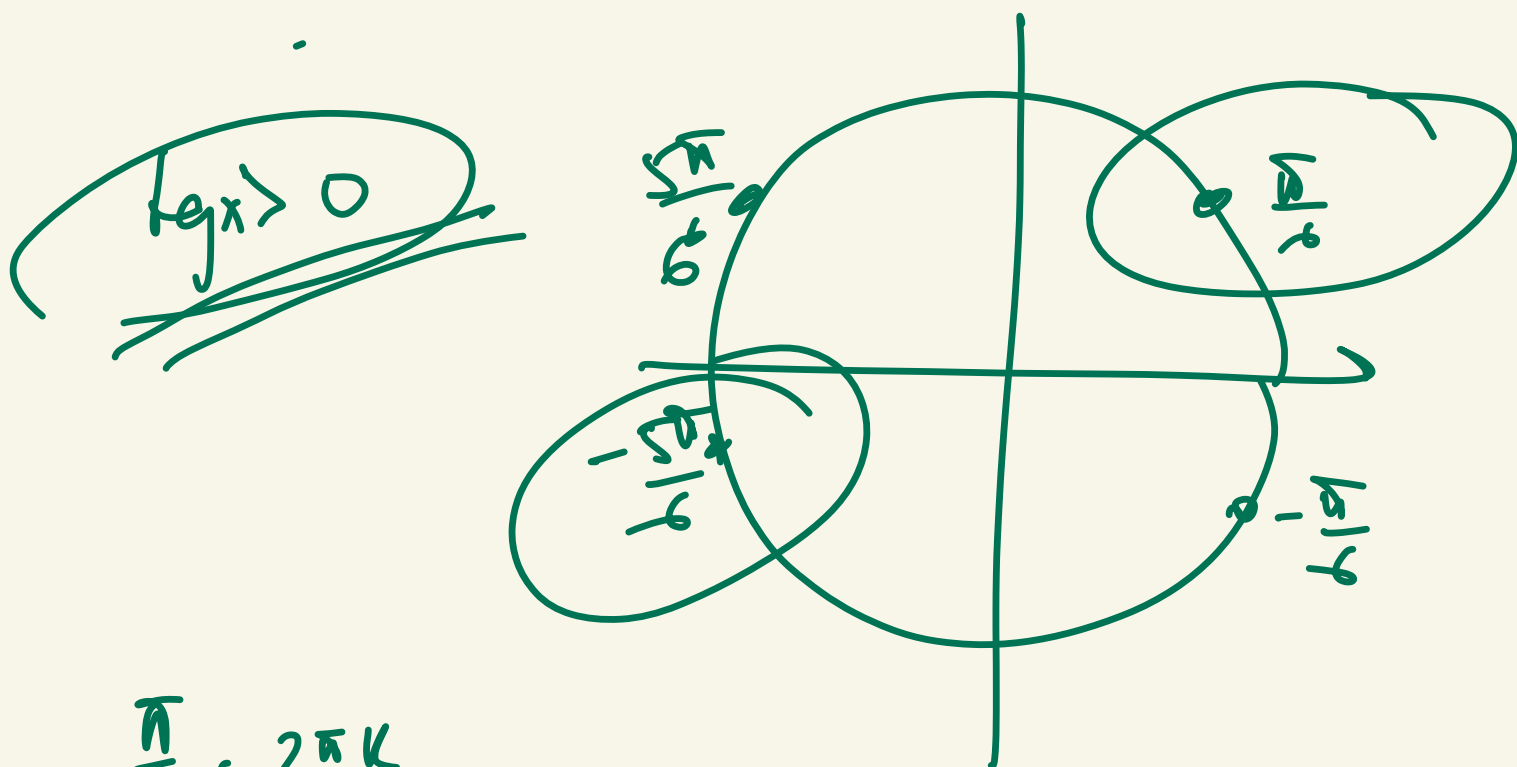
$$t_{\text{avg}} = \frac{165}{v_t + 4}$$

$$\log p = \frac{165}{v-4}$$

$$\log 2 + \log p = 22$$

$$\underline{22}^2 + \underline{3360} - \underline{352}$$

$$11v^2 - 165v - 176 = 0$$



$$\frac{\pi}{6} \in 2\pi\mathbb{K}$$

$$-\frac{5\pi}{6} + 2\pi\mathbb{K}$$

$$x^4 + 6x^3 + 7x^2 - 6x - 8 = 0$$

$$-8 \quad \begin{matrix} -1 & \cdot & 2 & -4 & -8 \\ 1 & 2 & 4 & 8 \end{matrix}$$

$$-1: \cancel{x} - 6 + \cancel{7} - 6 - \cancel{8} < 0$$

$$1: \underline{1} + \underline{6} + \underline{7} - \underline{6} - \underline{8} = 0$$

$$x = 1$$

$$\begin{array}{r} -x^4 + 6x^3 + 7x^2 - 6x - 8 \quad \frac{(x-1)}{x^3 + 7x^2 + 14x + 8} \\ x^4 - x^3 \\ \hline 7x^3 + 7x^2 \\ -7x^3 - 7x^2 \\ \hline 14x^2 - 6x \\ -14x^2 - 14x \\ \hline 8x - 8 \end{array}$$

$$(x-1)(x^3 + 7x^2 + 14x + 8) = 0$$

$$8 \quad \begin{matrix} \cancel{8} & \cancel{7} & \cancel{14} & \cancel{8} \\ \cancel{1} & \cancel{2} & \cancel{4} & \cancel{8} \end{matrix}$$

$$-2: \cancel{-8} + \cancel{28} - \cancel{28} + \cancel{8} = 0$$

$$- \frac{x^3 + 7x^2 + 14x + 8}{x^3 + 2x^2} \quad \frac{x+2}{x^2 + 5x + 4}$$

$$5x^2 + 14x$$

$$- 5x^2 + 10x$$

$$4x + 8$$

$$- 4x + 8$$

$$0$$

$$(x-1)(x+2)(x^2+5x+4) \leq 0$$

$$4x(x+3)(x+2)(x+3)$$

$$\frac{-5 \pm \sqrt{25-16}}{2} = \frac{-5 \pm 3}{2} \rightarrow \begin{matrix} 4 \\ -1 \end{matrix}$$

$$(x-1)(\cancel{x+2})(\cancel{x+3})(x+4)$$

$$4x(\cancel{x+3})(\cancel{x+2})(x+3)$$

$$\therefore (-3)$$

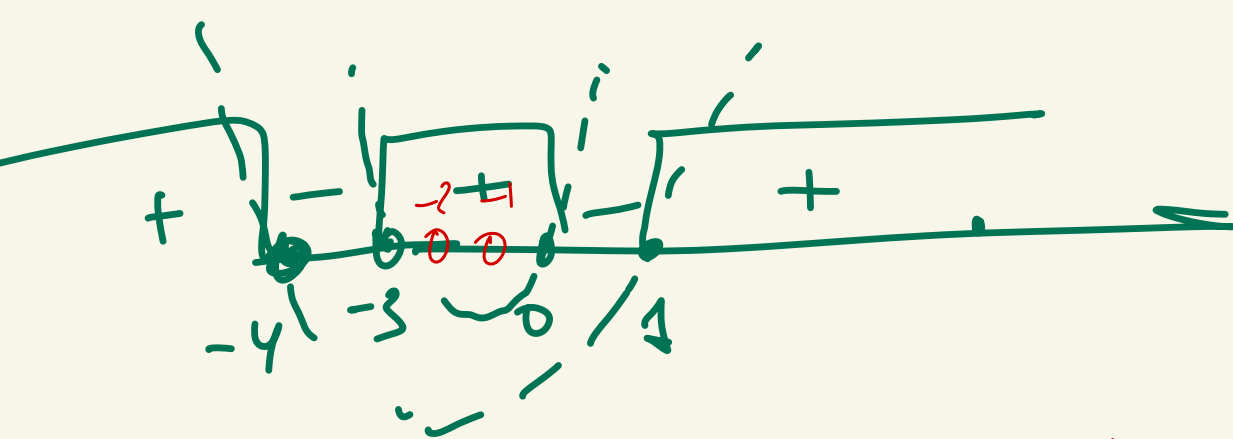
$$\neq 0$$

$$x \neq -1$$

$$x \neq -2$$

$$\frac{(x-1)(x+4)}{4x(x+3)}$$

$$\neq 0$$



$$x \in (-\infty : -4] \cup (-3 : -2) \cup (-2 : -1) \cup (-1 : 0) \cup [1 : +\infty)$$

$$\frac{1}{h(n+1)} = \frac{1}{n} - \frac{1}{n+1} = \frac{n+1-h}{n(n+1)} = \frac{1}{n(n+1)}$$

$$\frac{1}{x(x+1)} + \frac{1}{(x+1)(x+2)} + \frac{1}{(x+2)(x+3)} \leq \frac{3}{4}$$

$$\frac{1}{x} - \frac{1}{x+1} + \frac{1}{x+1} - \frac{1}{x+2} + \frac{1}{x+2} - \frac{1}{x+3} \leq \frac{3}{4}$$

$$x \neq -1$$

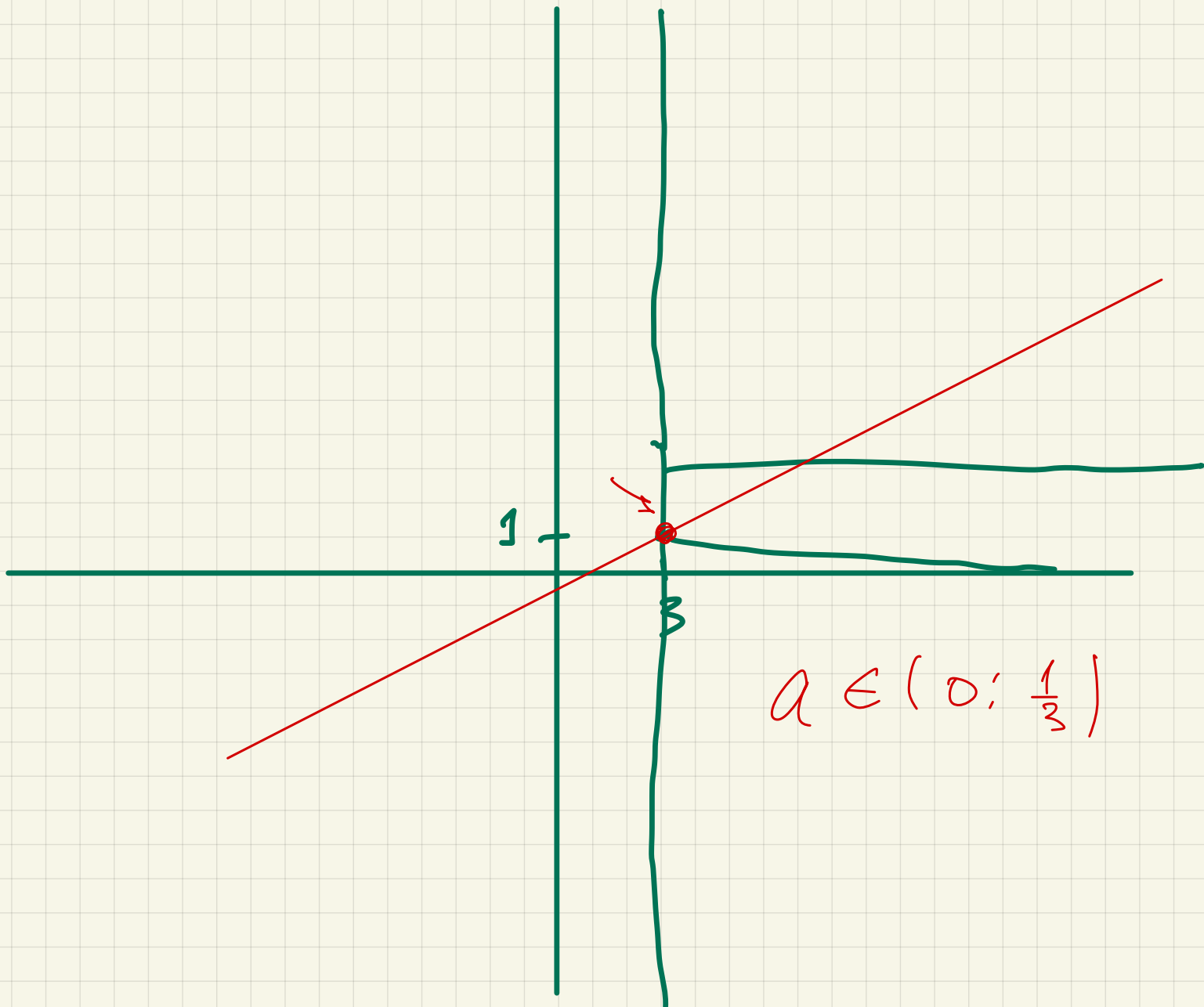
$$x \neq -2$$

$$\frac{1}{x} - \frac{1}{x+3} \leq \frac{3}{4}$$

$$\left\{ \begin{array}{l} x-3 = 0 \\ xy^2 - 3xy - 3y + 9 = 0 \\ x-3 \geq 0 \\ y = ax \end{array} \right.$$

$$\left\{ \begin{array}{l} (y-3)(xy-3) = 0 \\ x=3 \end{array} \right.$$

$$y = \frac{3}{x}$$



$$8a^2 + 24a + 4 \geq 0 ?$$

$$\rightarrow 4(2a^2 + 6a + 1) \geq 0$$

$$\frac{-6 \pm \sqrt{36 - 8}}{2 \cdot 2} = \frac{-6 \pm \sqrt{28}}{4}$$

$$= \frac{-6 \pm 2\sqrt{7}}{4}$$

$$= \frac{-3 \pm \sqrt{7}}{2}$$

$$\frac{-3 - \sqrt{7}}{2}$$

$$\frac{-3 + \sqrt{7}}{2}$$

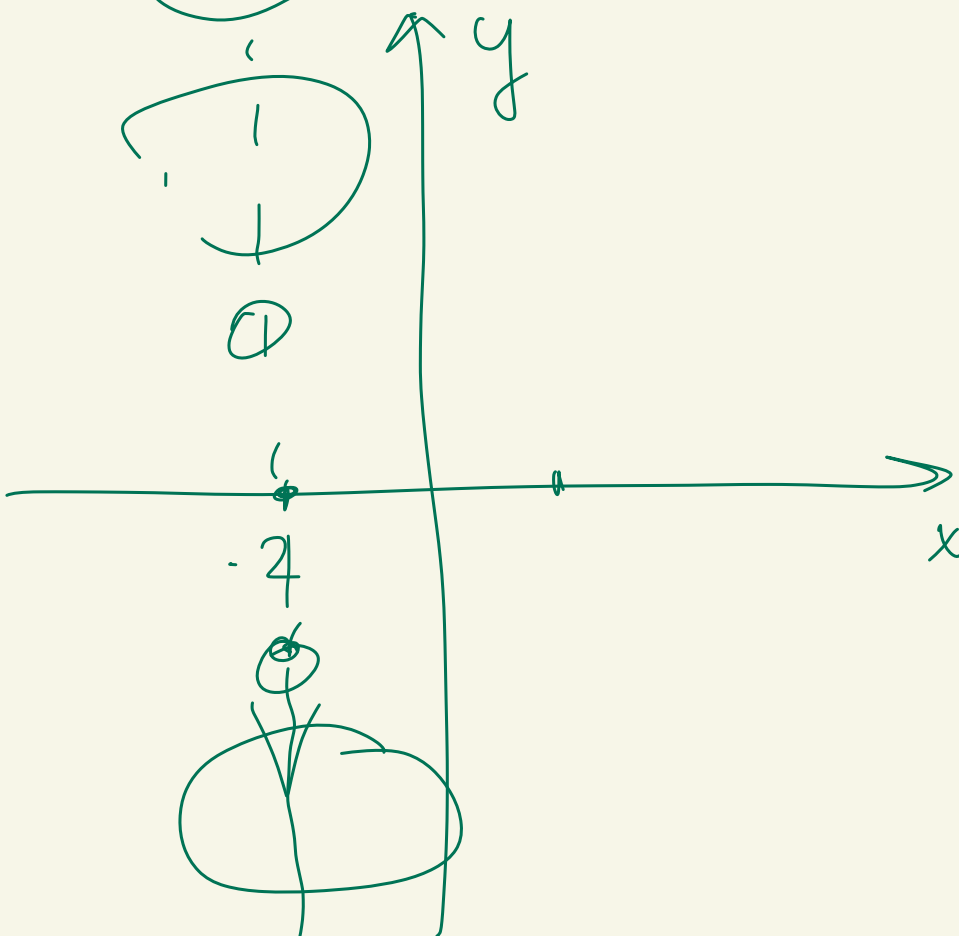
$\exists$ only 2

$$(-2; -3)$$

$$(-2; 0)$$

$a \uparrow$

a



$$(x+2)^2 + (y+3)^2 = 8 + 24 + 4 = 6^2$$

Hand-drawn geometric diagram on grid paper showing a coordinate system with x and y axes. A green circle is centered at $(-2, 0)$ with radius 2. A red circle is centered at $(-1, -3)$ with radius 3. The two circles are tangent at the point $(-3, 0)$. A dashed vertical line passes through the centers of both circles. Several lines with arrows are drawn, including a horizontal line $y=0$, a vertical line $x=-3$, and two diagonal lines passing through the centers of the circles. The diagram illustrates the geometric interpretation of the algebraic problem of finding the minimum value of the expression $(x^2 + 4x) + (a^2 - 6a + 9)$.

$$x^2 + 4x + 4 + 9 + 9a^2 + 18a - 8a^2 - 24a - 9 = 0$$

$$a = 1 \quad a = 5$$

$$\begin{cases} x^2 + y^2 + 2(2y - x)a = 1 + 2a - 4a^2 \\ x^2 + y^2 + 4(x - y)a = 4 + 4a - 7a^2 \end{cases}$$

$$x^2 - 2ax + y^2 + 4ay = 1 + 2a - 4a^2$$

$$x^2 - 2ax + \underbrace{a^2 - a^2} + y^2 + 4ay + \cancel{4a^2 - 4a^2}$$

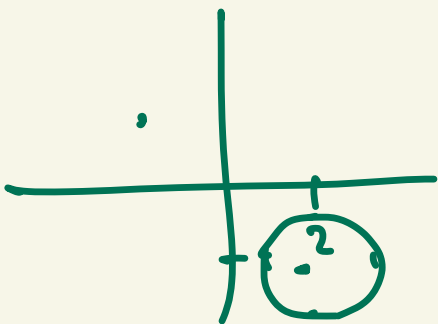
$$= 1 + 2a - \cancel{4a^2}$$

$$(x - a)^2 + (y + 2a)^2 = 1 + 2a + a^2 = (a + 1)^2$$

$$\begin{cases} (x - a)^2 + (y + 2a)^2 = |a + 1| & D_1(a; -2a) \\ (x + 2a)^2 + (y - 2a)^2 = |a + 2| & D_2(-2a; 2a) \end{cases}$$

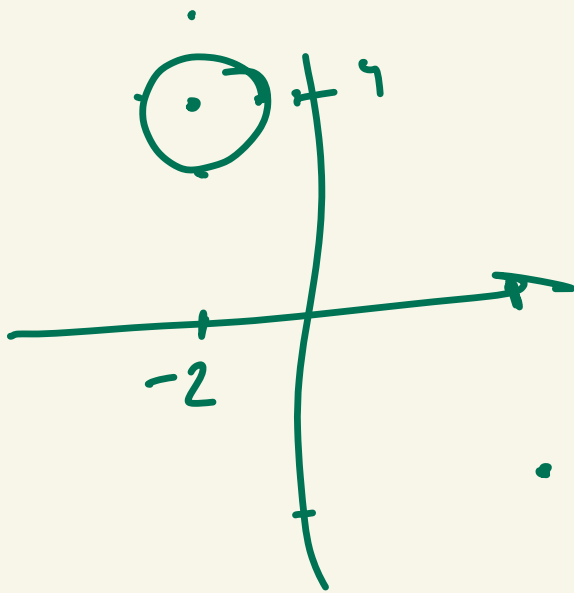
per $a = -1$

$$\begin{cases} (x + 1)^2 + (y + 2)^2 = 0 \\ (x - 2)^2 + (y + 2)^2 = 1 \end{cases}$$

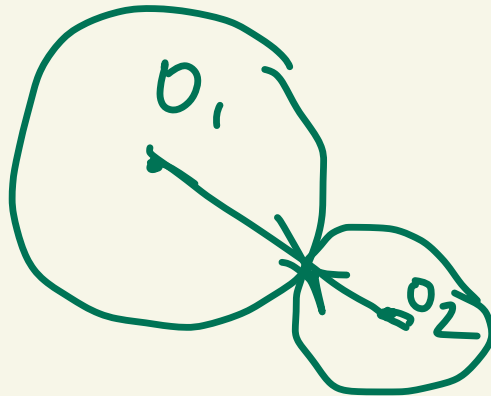


$$a = -2$$

$$\begin{cases} (x+2)^2 + (y-4)^2 = 1 \\ (x-4)^2 + (y+2)^2 = 0 \end{cases}$$



$$R \neq 0$$



$$1) |O_1, O_2| = R_1 + R_2$$

$$|O_1, O_2| = |R_1 - R_2|$$

$$\begin{aligned} O_1 & (a; -2a) \\ O_2 & (-2a; 2a) \end{aligned}$$

$$|O_1, O_2| = \sqrt{(a - (-2a))^2 + (-2a - 2a)^2}$$

$$\sqrt{9a^2 + 16a^2} = 5|a|$$

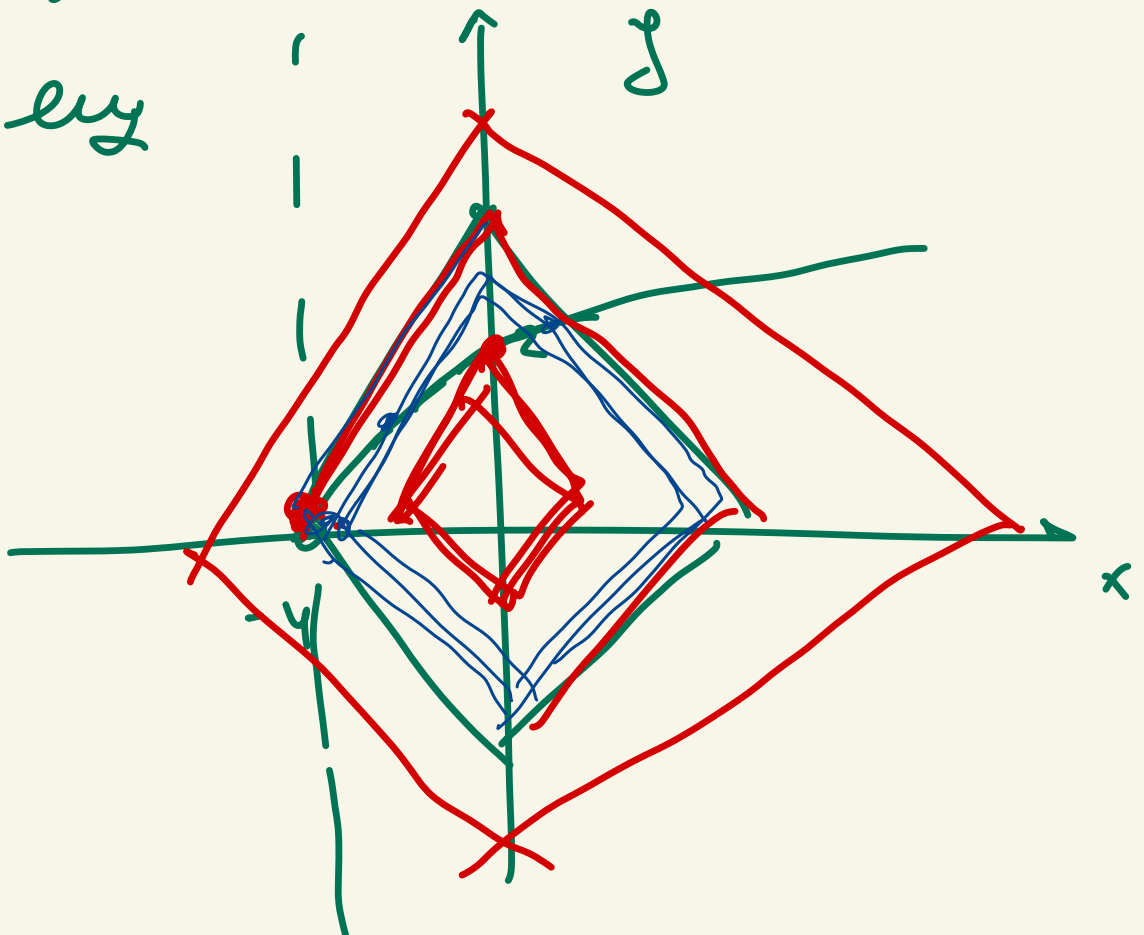
$$1) 5|a| = |1+a| + |2+a|$$

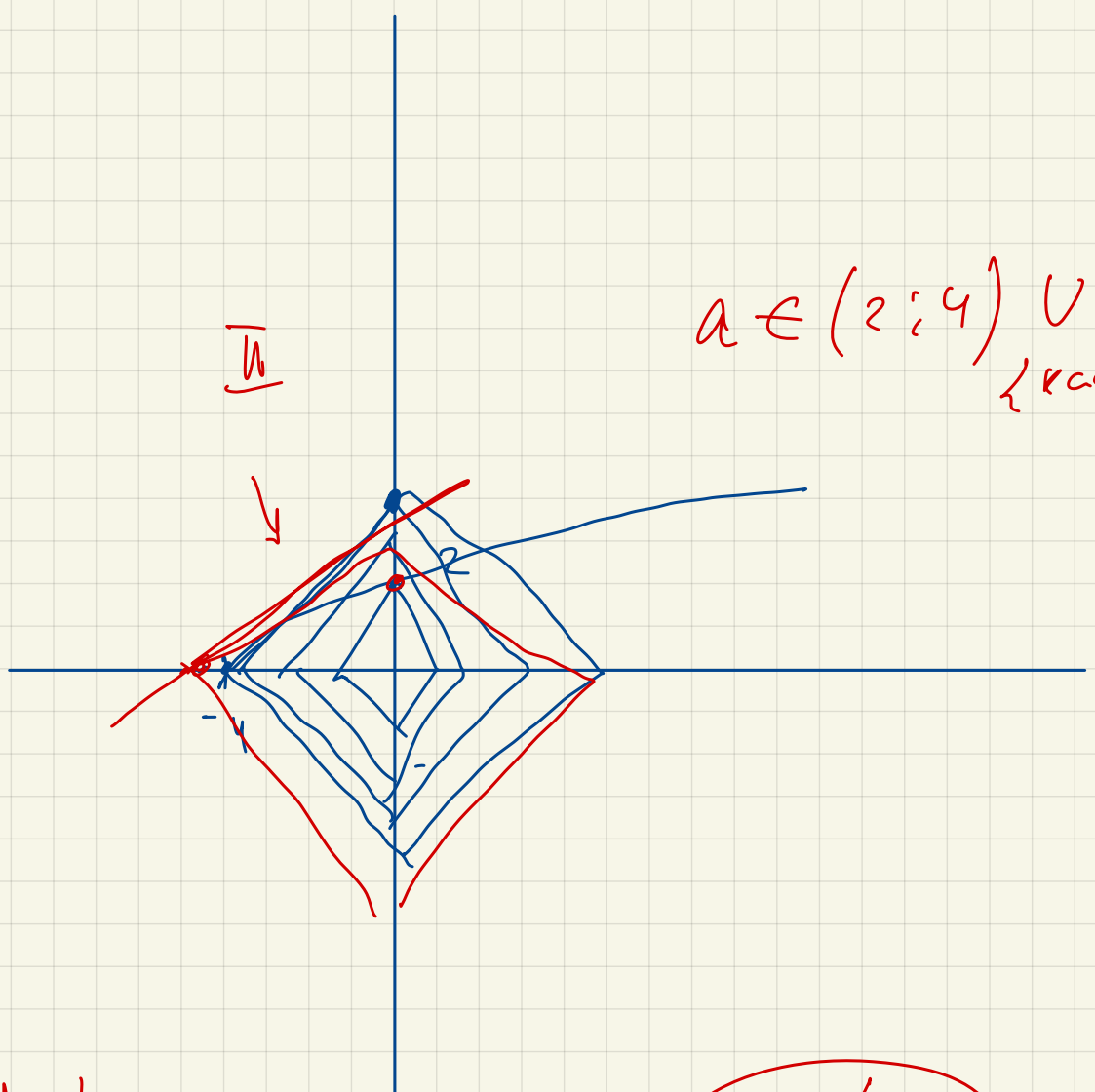
$$2) 5|a| = ||1+a| - |2+a||$$

0/3

$$\begin{cases} |x| + |y| = a \\ y = \sqrt{x+y} \end{cases} \quad x \geq -y$$

2 puz





$$a \in (2; 4) \cup \{x_{\text{center}}, \dots\}$$

$$|x| + |y| = a$$

$$|x| + \sqrt{x+4} = a$$

$$|x| + \sqrt{x+4} = a$$

$$x > 0$$

$$x \leq 0$$

$$\frac{1}{3}$$

